

Lab experiment – 3

```
33
34 def id3(rows, cols):
35     classes = set(r[-1] for r in rows)
36     if len(classes) == 1:
37         return classes.pop()
38     if not cols:
39         return Counter(r[-1] for r in rows).most_common(1)[0][0]
40     best = max(cols, key=lambda i: info_gain(rows, i))
41     tree = {attrs[best]: {}}
42     for v in sorted(set(r[best] for r in rows)):
43         subset = [r for r in rows if r[best] == v]
44         tree[attrs[best]][v] = id3(subset, [c for c in cols if c != best])
45     return tree
46
47 def classify(tree, sample):
48     while isinstance(tree, dict):
49         attr = list(tree)[0]
50         tree = tree[attr][sample[attr]]
51     return tree
52
53 tree = id3(data, [0, 1, 2, 3])
54 print("Decision Tree:")
55 print(tree)
56
57 new = {"Outlook": "Sunny", "Temp": "Cool", "Humidity": "High", "Wind": "Strong"}
58 print("\nNew sample:", new)
59 print("Prediction:", classify(tree, new))
```

input

Decision Tree:
{'Outlook': {'Overcast': 'Yes', 'Rain': {'Wind': {'Strong': 'No', 'Weak': 'Yes'}}, 'Sunny': {'Humidity': {'High': 'No', 'Normal': 'Yes'}}}}

New sample: {'Outlook': 'Sunny', 'Temp': 'Cool', 'Humidity': 'High', 'Wind': 'Strong'}
Prediction: No

lab experiment-4

```
1 import math, random
2
3 random.seed(1)
4 X = [[0, 0], [0, 1], [1, 0], [1, 1]]
5 Y = [0, 1, 1, 0]
6 H, lr = 4, 0.5
7 sig = lambda z: 1 / (1 + math.exp(-z))
8
9 W1 = [[random.uniform(-1, 1) for _ in range(H)] for _ in range(2)]
10 b1 = [0.0] * H
11 W2 = [random.uniform(-1, 1) for _ in range(H)]
12 b2 = 0.0
13
14 def forward(x):
15     h = [sig(sum(x[i] * W1[i][j] for i in range(2)) + b1[j]) for j in range(H)]
16     o = sig(sum(h[j] * W2[j] for j in range(H)) + b2)
17     return h, o
18
19 for e in range(10001):
20     loss = 0
21     for x, t in zip(X, Y):
22         h, o = forward(x)
23         loss += (t - o) ** 2 / 4
24         d2 = (o - t) * o * (1 - o)
25         d1 = [d2 * W2[j] * h[j] * (1 - h[j]) for j in range(H)]
26         for j in range(H):
27             W2[j] -= lr * d2 * h[j]
```

input

[0, 0]	0	0.009	0
[0, 1]	1	0.986	1
[1, 0]	1	0.986	1
[1, 1]	0	0.017	0

Accuracy: 100.0 %